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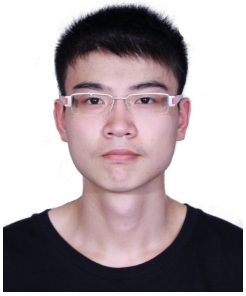

教育背景

广东工业大学, 机械工程, 博士 2016.09 - 2020.12

- **国家重点实验室**: 省部共建精密电子制造技术与装备国家重点实验室
- **学位论文**: 基于多物理场模型和机器学习模型的掩膜电射流加工表面微结构的理论与试验研究

广东工业大学, 机械工程, 硕士 2013.09 - 2016.07

- **国家重点实验室**: 省部共建精密电子制造技术与装备国家重点实验室
- **学位论文**: 电液束掩膜印制表面微织构仿真与实验研究
- **香港理工大学**, 微纳米精密实验室, 暑期课程 2013.07 - 2013.08

东莞城市学院, 机械设计制造及其自动化, 学士 2008.09 - 2012.07

职业经历

KU Leuven (2024 泰晤士世界大学排名 45), 副研究员 2024.03 - 至今

双学科:

- 制造系统与工程, 机械科学学院 (*Manufacturing Processes and Systems (MaPS)*, Department of Mechanical Engineering)
- 声明式语言与人工智能, 计算机学院 (*Declarative Languages and Artificial Intelligence (DTAI)*, Department of Computer Science)

KU Leuven (2024 泰晤士世界大学排名 45), 博士后 2021.03 - 2024.03

双学科:

- 制造系统与工程, 机械科学学院 (*MaPS*, Department of Mechanical Engineering)
- 声明式语言与人工智能, 计算机学院 (*DTAI*, Department of Computer Science)

KU Leuven (2024 泰晤士世界大学排名 45), 访问学者 2019.09 - 2020.09

广东省高校优秀青年科研人才国际培养计划

Tokyo University of Agriculture and Technology, 访问学者 2023.11

- **日本科学技术振兴机构（JST）**樱花科技交流计划资助

葡萄牙机械工业科学与工程研究院访问学者, 访问学者 2018.01 - 2018.01

瑞士洛桑联邦理工学院城市交通系统实验室, 访问学者 2017.07 - 2017.07

研究经历

- 研究方向:

- 人工智能与机器学习应用
- 多物理场建模
- 精密加工多传感在线监测与控制

- 非传统制造

- 项目经历 (博士毕业后)

GSF2025 **已申报, 2025 年底出结果**

- **项目描述** Global Seed Fund
- **关键合作伙伴** 浙江大学优青祝毅团队
- **职责** CO-PI

SYSTEMATIC, HEU2025 **已申报, 2026 年中出结果**

- **项目描述** 欧盟地平线项目 2025
- **关键合作伙伴** Ansys 等
- **职责** CO-PI

AI4AM, HEU2025 **已申报, 2026 年中出结果**

- **项目描述** 欧盟地平线项目 2025
- **关键合作伙伴** Gemini 等
- **职责** CO-PI

AutoCAM_SBO, Belgium **2024.10 - 至今**

- **项目描述** 比利时战略基础研究：基于人工智能的 CAD-to-CAM 加工方案规划与过程优化
- **项目总预算** ~275 万欧元
- **职责** 申报书起草者之一、研究骨干（1/1 DTAI 团队，1/2 MaPS 团队）及博士研究生的招聘与管理 基于生成式 AI 辅助的三维产品智能设计与制造工艺协同规划方法研究，结合产品的结构与性能要求、可用资源等约束，制定加工方案与并对工艺参数进行优化

MuSIC_SBO, Belgium **2022.10 - 至今**

- **项目描述** 比利时战略基础研究：多传感器融合的在线控制 (Multi-Sensor Fusion and In-process Control, MuSIC)
- **项目总预算** ~350 万欧元
- **职责** 申报书起草者之一，研究骨干（1/2 DTAI 团队） 基于人工智能的激光粉末床熔融（LPBF）和电弧增材制造（WAAM）过程缺陷检测与控制

AF3, Belgium **2022.10 - 至今**

- **项目描述** 自适应熔融沉积成型 (Adaptive Fused Filament Fabrication, AF3)
- **项目总预算** ~22 万欧元
- **职责** 研究骨干 (1/2 DTAI 团队) 基于数据-物理双驱动机制的线材打印工艺参数优化研究，利用热成像、声发射等多传感器数据进行实时监测与控制

CTO Action, Belgium **2022.06 - 2022.12**

- **项目描述** Flanders make CTO Action 2022
- **项目总预算** ~9.5 万欧元
- **职责** 研究人员 (2/2 MaPS 团队) 修订比利时工业 5.0 百科全书部分词条 (Triple helix, 人工智能、数字孪生、数字资产等部分).

ADAVI, Belgium **2021.03 - 2024.03**

- **项目描述** 比利时跨学科合作研究项目：基于主动视觉检测的异常检测 (Anomaly Detection by Active Visual Inspection, ADAVI)
- **项目总预算** ~300 万欧元
- **职责** 研究人员（2/2 MaPS 团队） 基于单目条纹投影的工业产品三维测量与表面缺陷检测

MicroMan, EU H2020 **2019.09-2020.09**

- **项目描述** 欧盟地平线 2020 计划：精密工程与微制造 (MicroMan)
- **项目总预算** ~450 万欧元
- **职责** 合作研究人员 (4/4 MaPS 团队) 精密工程领域的微电火花加工（EDM）/微电化学加工（ECM）研究

ProSurf, EU H2020 **2019.09-2020.09**

- **项目描述** 欧盟地平线 2020 计划：高性能功能表面结构的制造 (ProSurf)
- **项目总预算** ~475 万欧元
- **职责** 合作研究人员 (4/4 MaPS 团队) 基于电化学加工（ECM）的表面微结构制造研究

- 学术服务

- 为 **40+** 种学术期刊完成 **300+** 次审稿工作，包括 EAAI、ESWA、JMP 等中科院一区 TOP 期刊
- 担任 International Journal of Industrial and Manufacturing Systems Engineering 编委
- 担任 Symmetry (Q1: General Mathematics) 特刊学术编辑
- 自 2022 年起担任美国机械工程师学会（ASME）会员
- 自 2022 年起担任电气与电子工程师学会（IEEE）会员
- 自 2021 年起担任比利时弗兰德斯制造 (Flanders Make) 协会会员

学术发表

期刊论文

一作/通讯

1. **M. Wu**, J. Liu, J. He, X. Chen, and Z. Guo*, Fabrication of surface microstructures by mask electrolyte jet machining, International Journal of Machine Tools and Manufacture 148 no. ARTN 103471, (2020), DOI: 10.1016/j.ijmachtools.2019.103471, **中科院一区 TOP IF=18.8**
2. **M. Wu**, Z. Yao, P. Karsmakers, M. Verbeke, B. Gorissen, and D. Reynaerts*, Data-driven models with physical interpretability for real-time cavity profile prediction in electrochemical machining processes, Engineering Applications of Artificial Intelligence 160, (2025), 111807, DOI: 10.1016/j.engappai.2025.111807, **中科院一区 TOP, IF=8**
3. Z. Yao, J. Ge, P. Cao, **M. Wu***, J. Qian, Y. Li, and D. Reynaerts*, Advancements in process monitoring and quality control for electrical discharge machining: A comprehensive review, Journal of Materials Processing Technology 345, (2025), 119081, DOI: 10.1016/j.jmatprotec.2025.119081, **中科院一区 TOP IF=7.5**
4. J. Ge, Z. Yao, **M. Wu***, J. H. S. Almeida, Y. Jin, and D. Sun, Tackling data scarcity in machine learning-based cfrp drilling performance prediction through a broad learning system with virtual sample generation (bls-vsg), Composites Part B: Engineering, (2025), 112701, DOI: 10.1016/j.compositesb.2025.112701, **中科院一区 TOP IF=14.2**
5. Z. Yao, **M. Wu***, J. Qian, and D. Reynaerts*, Intelligent discharge state detection in micro-edm process with cost-effective radio frequency (rf) radiation: Integrating machine learning and interpretable ai, Expert Systems with Applications 291, (2025), 128607, DOI: 10.1016/j.eswa.2025.128607, **中科院一区 TOP**
6. **M. Wu**, M. Hazak Arshad, K. Kumar Saxena, D. Reynaerts, Z. Guo, and J. Liu*, Experimental and numerical investigations on fabrication of surface microstructures using mask electrolyte jet machining and duckbill nozzle,

Journal of Manufacturing Science and Engineering -Transactions of the ASME 145 no. 5, (2023), DOI: 10.1115/1.4056570, 中科院二区

7. Y. Lin, Y. Lin, H. Wu*, and **M. Wu***, The enhance-fuse-align principle: A new architectural blueprint for robust object detection, with application to x-ray security, Sensors 25 no. 21, (2025), 6603, DOI: 10.3390/s25216603
8. **M. Wu**, Z. Guo, J. Qian, and D. Reynaerts*, Multi-ion-based modelling and experimental investigations on consistent and high-throughput generation of a micro cavity array by mask electrolyte jet machining, Micromachines 13 no. 12, (2022), DOI: 10.3390/mi13122165, 中科院三区
9. **M. Wu**, K. Kumar Saxena, Z. Guo, J. Qian, and D. Reynaerts, Fast fabrication of complex surficial micro-features using sequential lithography and jet electrochemical machining, Micromachines 11 no. 10, (2020), DOI: 10.3390/mi11100948, 中科院三区
10. X. Zhang, J. Wang, J. Wei, X. Yuan, and **M. Wu***, A review of non-fully supervised deep learning for medical image segmentation, Information 16 no. 6, (2025), DOI: 10.3390/info16060433, 中科院四区
11. Y. Shi, Y. He, J. Zha*, B. Chen, C. Shi, and **M. Wu***, Design and manufacture of a flexible adaptive fixture for precision grinding of thin-walled bearing rings, Journal of Manufacturing and Materials Processing 9 no. 5, (2025), 139, DOI: 10.3390/jmmp9050139, 中科院四区
12. **M, Wu**, J. Zhang, R. Abts, E. Ferraris, and M. Verbeke*, Deep learning-based characterization of fused filament fabrication from temporal thermal data, Procedia CIRP 133, (2025), 573–578, DOI: 10.1016/j.procir.2025.02.098, EI
13. **M, Wu**, S. Shukla, B. Vrancken, M. Verbeke, and P. Karsmakers*, Data-driven approach to identify acoustic emission source motion and positioning effects in laser powder bed fusion with frequency analysis, Procedia CIRP 133, (2025), 531–536, DOI: 10.1016/j.procir.2025.02.091, EI
14. **M, Wu**, Z. Yao, R. Abts, P. Karsmakers, M. Verbeke, and D. Reynaerts*, A threshold-free and label-free pipeline for adaptive pulse classification in electrical discharge machining, Procedia CIRP 133, (2025), 692–697, DOI: 10.1016/j.procir.2025.02.118, EI
15. **M, Wu**, Z. Yao, M. Verbeke, P. Karsmakers, and D. Reynearts*, Geometrical feature classification in electrical discharge machining using in-process monitoring and machine learning, Procedia CIRP 137, (2025), 462–467, DOI: 10.1016/j.procir.2025.07.010, EI
16. **M, Wu**, M. Arshad, K. Saxena, J. Qian, and D. Reynaerts*, Profile prediction in ecm using machine learning, Procedia CIRP 113, (2022), 410–416, DOI: 10.1016/j.procir.2022.09.192, EI
17. **M, Wu**, Z. Guo, J. Qian, and D. Reynaerts*, Fabrication of surface micro letters by electrolyte jet mask machining, Procedia CIRP 95, (2020), 827–832, DOI: 10.1016/j.procir.2020.02.261, EI
18. **M, Wu**, Z. Guo*, J. He, and X. Chen, Modeling and simulation of the material removal process in electrolyte jet machining of mass transfer in convection and electric migration, Procedia CIRP 68, (2018), 488–492, DOI: 10.1016/j.procir.2017.12.079, EI
19. **M. Wu**, Z. Guo, T. Sun, H. Luo, and Q. Song, Investigation on pulse electrocheimical machining by coated electrode, Modern Manufacturing Engineering 2015 no. 11, (2015), 11–14, DOI: 10.3969/j.issn.1671-3133.2015.11.004

合作

20. L. Ye, C. Guo, **M. Wu**, J. Qian, N. Yu, and D. Reynaerts, Interpretable deep temporal neural networks for in-situ monitoring under varying conditions in micro-electrical discharge machining, Journal of Manufacturing Systems 81, (2025), 222–237, DOI: 10.1016/j.jmsy.2025.05.007, 中科院一区 TOP IF=12.3
21. B. Chen, J. Zha*, Z. Cai, and **M. Wu**, Predictive modelling of surface roughness in precision grinding based on hybrid algorithm, CIRP Journal of Manufacturing Science and Technology 59, (2025), 1–17, DOI: 10.1016/j.cirpj.2025.02.004, 中科院二区
22. Y. Zhu*, D. Li, C. Zhang, F. Lin, **M. Wu**, and Y. Chen, Abrasive flow machining of laser powder bed fusion fabricated mini-channels: Modelling and verification, Journal of Manufacturing Processes 141, (2025), 36–47, DOI: 10.1016/j.jmapro.2025.02.073, 中科院一区 TOP 合作-浙大优青，ASME 会士
23. J. Ge, J. Zhang, M. Xu, **M. Wu**, Z. Yao, G. Fu, W. Zhang, B. G. Falzon, G. Catalanotti, J. Xu, Y. Jin, and D. Sun, Data-driven CFRP machining performance prediction and optimization: Advances, challenges and future prospects, Thin-Walled Structures, (2025), 113721, DOI: 10.1016/j.tws.2025.113721
24. H. Ding, Y. Wang, Q. Tang, Y. Sun, **M. Wu**, W. Wang, J. Bai*, H. Xie, H. Yang, C. Zhang, and Y. Zhu*, Enhancing cavitation erosion resistance of 316l stainless steel through simultaneous microstructure modification during l-pbf, Tribology Transactions, (2025), 1–16, DOI: 10.1080/10402004.2025.2476013,

四区合作-浙大院士优青

25. J. Liang, A. Chen, **M. Wu**, X. Tang, H. Feng, J. Liu*, and G. Xie*, A shellfish-inspired bionic microstructure design for biological implants: Enhancing protection of antibacterial silver-loaded coatings and promoting osseointegration, Journal of the Mechanical Behavior of Biomedical Materials, (2025), 106963, DOI: 10.1016/j.jmbbm.2025.106963, 中科院三区
26. D. Li, T. Zhang, **M. Wu**, W. Wang, H. Ding, F. Lin, and Y. Zhu, Cavitation intensity mechanism in hydrodynamic cavitation abrasive finishing, Journal of Manufacturing Science and Engineering -Transactions of the ASME, (2025), 1–17, DOI: 10.1115/1.4068753, 中科院三区，合作-浙大优青
27. G. Li, P. Hu, Y. Fan, **M. Wu**, M. Chen, and T. Yu, Anisotropic tensile properties and high-cycle fatigue behavior of ti6al4v fabricated by single and dual laser-powder bed fusion, Materials Science and Engineering: A, (2025), 148640, DOI: 10.1016/j.msea.2025.148640, **中科院二区**合作-哈工大优青
28. R. Xu, Y. Zhu, J. Wu, P. Huang, **M. Wu**, W. Wang, C. Zhang*, and H. Yang, Hybrid high-temperature wear mechanisms of additive manufactured ti-6al-4v alloy, Tribology International 205, (2025), 110559, DOI: 10.1016/j.triboint.2025.110559, **中科院一区 TOP** 合作-浙大院士优青
29. J. He, W. Chen, J. Wang, **M. Wu**, L. Zhou, R. Chen, and H. Liang, Exploring electrochemical direct writing machining of patterned microstructures on zr702 with polyacrylamide polymer electrolyte, Micromachines 15 no. 9, (2024), DOI: 10.3390/mi15091074, 中科院三区
30. Z. Yao, L. Ye, **M. Wu**, J. Qian, and D. Reynaerts, Prediction of crater morphology and its application for enhancing dimensional accuracy in micro-edm, Journal of Intelligent Manufacturing, (2025), DOI: 10.1007/s10845-024-02430-2, 中科院二区 IF=5.9
31. Z. Yao, J. Shen, **M. Wu**, D. Zhang, and M. Luo, Position-dependent milling process monitoring and surface roughness prediction for complex thin-walled blade component, Mechanical Systems and Signal Processing 198 no. ARTN 110439, (2023), DOI: 10.1016/j.ymssp.2023.110439, 中科院一区 TOP IF=8 合作-西工大优青
32. Z. Yao, **M, Wu**, J. Qian, and D. Reynaerts*, Tool wear evaluation and its evolution in micro-edm using radio frequency (rf) signals, Procedia CIRP 137, (2025), 158–163, DOI: 10.1016/j.procir.2025.07.006, EI
33. K. Saxena, M. Arshad, **M, Wu**, J. Qian, and D. Reynaerts*, A ‘virtual sensing’ framework for µecm/hybrid laser-µecm for monitoring interelectrode gap conditions, Procedia CIRP 113, (2022), 445–452, DOI: 10.1016/j.procir.2022.09.198, EI
34. M. Arshad, **M, Wu**, K. Saxena, J. Qian, S. Huang, and D. Reynaerts*, Experimental investigations into machining characteristics of niobium carbide cermet with µecm / hybrid laser-µecm, Procedia CIRP 113, (2022), 385–391, DOI: 10.1016/j.procir.2022.09.188, EI
35. H. Zhuang, Z. Guo, **M. Wu**, and J. Wang, Experimental study on electrode machining of micro-pit arrays by scanning jet mask, Machinery Design & Manufacture no. 3, (2022), 139–143, DOI: 10.3969/j.issn.1001-3997.2022.03.028
36. M. Bellotti, **M. Wu**, J. Qian, and D. Reynaerts, Tool wear and material removal predictions in micro-edm drilling: Advantages of data-driven approaches, Applied Sciences 10 no. 18, (2020), DOI: 10.3390/app10186357, 中科院三区
37. J. Liu, Q. Huang, **M. Wu**, Z. Zou, Z. Lin, Z. Guo, J. He*, and X. Chen*, Electrochemical discharge grinding of metal matrix composites using shaped abrasive tools formed by sintered bronze/diamond, Science And Engineering Of Composite Materials 27 no. 1, (2020), 346–358, DOI: 10.1515/secm-2020-0038, 中科院四区
38. H. Zhuang, Z. Guo, J. Wang, and **M. Wu**, Study of injection molding microstructure of stainless steel based on jet mask electrochemical machining, Journal of Guangdong University of Technology 3, (2020), 70–74, DOI: 10.12052/gdutxb.190101
39. L. Wen, Z. Guo, C. Chen, X. Chen, **M. Wu**, and Y. Deng, The simulation and experimental study of the mask jet-ecm machining inner wall micro-structure, Modern Manufacturing Engineering, (2019), 1–4, DOI: 10.16731/j.cnki.1671-3133.2019.06.001
40. X. Chen, B. Dong, C. Zhang, **M. Wu**, and Z. Guo, Jet electrochemical machining of micro dimples with conductive mask, Journal of Materials Processing Technology 257, (2018), 101–111, DOI: 10.1016/j.jmatprotec.2018.02.035, 中科院二区
41. J. He, **M. Wu**, R. Zhang, J. Liu, Y. Deng, and Z. Guo, A one-step hot-embossing process for fabricating a channel with superhydrophobic inner walls, Journal of Manufacturing Processes 36, (2018), 351–359, DOI: 10.1016/j.jmapro.2018.10.036, 中科院一区
42. Y. Qin, J. He*, F. Yang, Z. Zhang, Z. Yuan, and **M. Wu**, Extraordinary optical transmission properties of a novel

bi-layered plasmonic nanostructure array, Optik 174, (2018), 684–691, DOI: 10.1016/j.ijleo.2018.08.073, 中科院三区

43. R. Zhang, Z. Guo, S. Jiang, **M. Wu**, H. Luo, and G. Wang, Preparation of super-hydrophobic surface of sic/2024al composite through chemical etching and its properties, Corrosion Science and Protection Technology 30 no. 1, (2018), 61–66, DOI: 10.11903/1002.6495.2017.068, EI
44. C. Zhou, Y. Zhang, G. Liu, **M. Wu**, and Y. Yao, Experimental study on cathode optimized ecm-shaped hole, Electromachining & Mould, (2016), 20–32, DOI: 10.3969/j.issn.1009-279X.2016.03.006
45. T. Sun, Z. Yu, **M. Wu**, and G. Wang, Development and application of wall surface micro-machining technology, Modern Manufacturing Engineering 2015 no. 12, (2015), 147–152, DOI: 10.3969/j.issn.1671-3133.2015.12.030
46. T. Sun, Z. Yu, **M. Wu**, G. Wang, and L. Cheng, Simulation and experimental study on flow field of the spiral electrode electrochemical micro-machining, Electromachining & Mould, (2015), 30–33, DOI: 10.3969/j.issn.1009-279X.2015.01.008
47. Y. Tang, H. Luo, **M. Wu**, T. Sun, and X. Yin, Developments and applications of jet electrochemical machining, Electromachining & Mould, (2015), 11–15, DOI: 10.3969/j.issn.1009-279X.2015.z1.003

会议

1. **M, Wu**, S. Shukla, B. Vrancken, M. Verbeke, and P. Karsmakers*, *Data-driven approche to identify disortion of acoustic signals by process contextural information in lpbf*, Leuven, Belgium, (2025)
2. **M, Wu**, J. Zhang, R. Abts, E. Ferraris, and M. Verbeke*, *Deep learning-based characterization of fused filament fabrication fromtemporal thermal data*, Leuven, Belgium, (2025)
3. **M, Wu**, S. Kushwaha, Q. Jun, D. Reynaerts, and M. Verbeke*, *Linking 3d models and manufacturing operations through point-based labelling for cam knowledge digitalization*, Leuven, Belgium, (2025)
4. **M, Wu**, J. Zhang, R. Abts, E. Ferraris, and M. Verbeke*, *Classification of fused filament fabrication stability from temporal infrared thermal data using x3d*, Leuven, Belgium, (2025)
5. **M, Wu**, J. Qian, M. Verbeke, P. Karsmakers, and D. Reynaerts*, *Surface inspection of highly reflective rotary products using fringe projection with high dynamic range*, Kortrijk, (2024)
6. **M, Wu**, Z. Yao, J. Qian, M. Verbeke, P. Karsmakers, and D. Reynaerts*, *Enhanced pulse classification in electric discharge machining through semi-supervised learning with dynamic time warping*, Kortrijk, (2024)
7. **M, Wu**, S. Kushwaha, J. Qian, D. Reynaerts, M. Verbeke, and P. Karsmakers*, *Representation of machining state and optimization of rough milling efficiency using machine learning*, Antwerp Expo, Belgium, (2023)
8. **M, Wu**, J. Qian, and D. Reynaerts*, *A machine learning model for profile precition in ecm*, Gent, Belgium, (2023)
9. **M, Wu**, Z. Guo*, X. Chen, and C. Zhang, "Investigation on pulse electrochemical machining of aluminum alloy 7075," in *2017 The 8th International Conference on Mechanical and Aerospace Engineering*, Prague, Czech Republic, (2017)
10. **M, Wu**, M. Arshad, K. Saxena, J. Qian, and D. Reynaerts*, "Data-driven framework of ecm: A machine learning model for profile prediction," in *Proceedings of 17th International Symposium on Electrochemical Machining Technology (INSECT 2021)*, Leuven, Belgium (hosted virtually due to Covid-19), KU Leuven - Departement Werktuigkunde, (2021), 231–240, ISBN: 978-9-08289-313-7
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专利及成果转化

- 专利

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- 其他

1. 基于深度学习的网页版电化学加工实时工艺参数预测：mingwucn.github.io/SmartECM/
2. 电火花加工自适应脉冲分类桌面应用，已作为开源软件发布
3. 面向高反射率回转体表面的单目结构光桌面软件，受 KU Leuven 产权保护

教学活动

- **博士生导师**: 鲁汶大学博士 (2025): Al-assisted CAD-to-CAM optimization for Machining Processes
- **客座讲师**: 鲁汶大学 2024 冬季研究生课程 *AI in Manufacturing*
- **硕士生导师**: 鲁汶大学硕士论文 (2025) Co-pilot for function-driven product design: automated generation of 3D designs tailored to specific functional requirements
- **硕士生导师**: 鲁汶大学硕士论文 (2022): Experimental investigation of μ-ECM/μ-LECM process with machine learning
- **志愿者讲师**: 2021 年贵州 ICARE 在线课程公益项目（5 月-6 月）